

```
#include <Servo.h>
```

```
Servo horizontal;
```

```
int servoh = 90; // Horizontal  
servo start position
```

```
Servo vertical;
```

```
int servov = 90; // Vertical servo  
start position
```

```
int ldrlt = A0; // top left
```

```
int ldrrt = A1; // top right
```

```
int ldrld = A2; // down left
```

```
int ldrrd = A3; // down right
```

```
void setup() {  
    Serial.begin(9600); // Isse Serial  
    Monitor chalega  
    horizontal.attach(10); //  
    Horizontal servo pin  
    vertical.attach(11); // Vertical  
    servo pin  
  
    horizontal.write(servoh);  
    vertical.write(servov);  
    delay(2000);  
}
```

```
void loop() {  
    int It = analogRead(ldrIt); // top
```

left

```
int rt = analogRead(ldrrt); // top
```

right

```
int ld = analogRead(ldrld); //
```

down left

```
int rd = analogRead(ldrrd); //
```

down right

```
Serial.print("TL: ");
```

```
Serial.print(lt); Serial.print(" | ");
```

```
Serial.print("TR: ");
```

```
Serial.print(rt); Serial.print(" | ");
```

```
Serial.print("DL: ");
```

```
Serial.print(ld); Serial.print(" | ");
```

```
Serial.print("DR: ");
```

```
Serial.println(rd);
```

```
int dtime = 10;
```

```
int tol = 50;
```

```
int avt = (lt + rt) / 2; // average  
top
```

```
int avd = (ld + rd) / 2; // average  
down
```

```
int avl = (lt + ld) / 2; // average  
left
```

```
int avr = (rt + rd) / 2; // average  
right
```

```
int dvert = avt - avd; // check  
difference up/down
```

```
int dhoriz = avl - avr; // check  
difference left/right
```

```
// Vertical movement  
if (-1*tol > dvert || dvert > tol) {  
    if (avt > avd) {  
        servov = ++servov;  
        if (servov > 180) { servov =  
180; }  
    } else if (avt < avd) {  
        servov = --servov;  
        if (servov < 0) { servov = 0; }  
    }  
    vertical.write(servov);  
}
```

```
// Horizontal movement
if (-1*tol > dhoriz || dhoriz > tol) {
    if (avl > avr) {
        servoh = --servoh;
        if (servoh < 0) { servoh = 0; }
    } else if (avl < avr) {
        servoh = ++servoh;
        if (servoh > 180) { servoh =
180; }
    }
    horizontal.write(servoh);
}

delay(dtime);
}
```

